

Ope Edet

+234 802 100 2168 | opeotoabasi@gmail.com | github.com/opeotoabasi

Professional Summary

Second-year Electrical and Electronics Engineering student at Obafemi Awolowo University building Linux systems software in C/C++, including a custom Unix shell with process and signal control, a LISP interpreter, and a concurrent price-time priority limit order book and matching engine. 200+ problems solved across Codeforces and LeetCode. Seeking an infrastructure / systems software engineering internship focused on reliable, high-performance production systems.

Education

Obafemi Awolowo University (OAU)

Ile-Ife, NG

Bachelor of Science in Electrical and Electronics Engineering

Aug. 2024 – Present (Expected

Nov. 2030)

Relevant Coursework: Embedded Systems (Arduino), Statistics

Projects

Custom Unix Shell | C, Linux

- Built a Unix-like command shell from scratch to gain a first-principles understanding of how Linux manages processes and I/O.
- Implemented the prompt loop, command parsing, and built-in commands, with process creation and control using fork, exec, and wait, plus input/output redirection via file descriptors.
- Currently extending the shell with signal handling and job control to support background processes and process groups.

Limit Order Book / Matching Engine | C++, CMake

- Designed and built a price-time priority limit order book and matching engine in C++ to deepen practical fluency with the low-latency, data-structure-heavy systems used in financial technology.
- Implemented core order book logic supporting limit and market order matching and order cancellation, in a multi-file CMake structure separating interface, implementation, and tests.
- Extended the engine with a concurrent producer-consumer order queue, a TCP networking layer, and SQLite persistence.

LISP Interpreter | C, GDB

- Implemented a Lisp interpreter in C, building a grammar and parser with the mpc parser combinator library to construct and evaluate abstract syntax trees (ASTs).
- Designed lval structures to represent and evaluate expressions, including manual memory management and Q-expressions (unevaluated expressions) as data-manipulation primitives.

Leadership & Achievements

- **1st Place, Circom** – solo circuit assembly competition, SPAC IEEE OAU (Student Activities Committee, IEEE OAU Student Branch).
- **President, JETS Club** (Junior Engineers, Technicians and Scientists) – took full ownership of club activities and project teams, represented the school at a state-level tech fair, and grew the club's population by over 200%.

Technical Skills

Systems & Linux: process control, signals, I/O redirection, file descriptors, memory management, concurrency, parsing (ASTs); TCP networking

Languages: C, C++ (primary), Python

Tools: Linux CLI, Git, GDB, CMake, SQLite

Data Structures & Algorithms: hash maps, trees, priority queues, graphs; 200+ problems solved on Codeforces and LeetCode